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Dark Matter from Baryon Density

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ABSTRACT

We present a novel empirical equation for the dark matter derived only from galactic (gas) kinematics. It is defined simply and entirely by the equation $\rho_{\rm dm} = m_e\alpha^2/a_0 d^2$, where d is the mean particle spacing in normal matter, a_0 is the Bohr radius, α and m_e are the usual. The meaning of the relationship is left open, rather we evaluate its applicability, by optimizing and testing this relation against the complete Spitzer Photometry and Accurate Rotation Curves (SPARC) database, with over 3000 data points across 175 morphologically diverse galaxies. The scaling law varies the dark mass density with the $2/3$ power of the matter density. This results in dark matter starting to be substantial when baryons are about a millimetre apart. The model generates rotation curves that are very similar to MOND, but has the advantage that it generates massive dark matter. With a simple density function for the circumgalactic medium (CGM) gas, it simultaneously produces appropriate dark matter halo masses as well.

Key words: dark matter – galaxies: kinematics and dynamics – galaxies: haloes

1 INTRODUCTION

1.1 History

For over four decades, the standard cosmological paradigm (ΛCDM) has successfully explained large-scale structure formation, cosmic microwave background anisotropies, and cosmic shear observations by invoking a dominant component of cold dark matter. However, at galactic and sub-galactic scales, the nature of this missing mass field remains one of the most persistent crises in modern physics. Despite intensive efforts spanning astronomical observations, deep underground direct-detection experiments, and high-energy collider searches, particle dark matter candidates—ranging from Weakly Interacting Massive Particles (WIMPs) to ultra-light axions—have eluded detection across nearly thirty orders of magnitude in predicted mass scales. Even assuming any (perhaps undetectable) particle like dark matter, problems persist. For example, modelling the dark matter distribution in galaxies, where Sancisi's Law(Sancisi 2004) (Lelli et al. 2016), (Lelli et al. 2017)and the puzzling excellent performance of the MOND framework indicate that a conventional particle dark matter model may never work.

1.2 Current Approaches

There are currently two primary paths of investigation in galactic dynamics. The first relies on multi-parameter empirical dark matter density profiles, such as the Navarro-Frenk-White (NFW)(Navarro et al. 1997) or Burkert profiles(Burkert 1995), which are fitted to individual galactic rotation curves. While these profiles effectively capture overall kinematic features, they generally treat parameters like scale radius (r_s) and central density (ρ_0) as free, independent variables unique to each galaxy. This introduces a large parameter space that requires individual, galaxy-by-galaxy 'tuning.' The second path, Modified Newtonian Dynamics (MOND), abandons the dark matter hypothesis entirely, accounting for anomalous galactic rotation curves by introducing a universal acceleration constant ($a_0 \approx 1.2 \times 10^{-10}$ m s^{-2}) below which Newtonian gravity breaks down. MOND's striking empirical success, particularly in explaining the Radial Acceleration Relation (RAR) and the Baryonic Tully-Fisher Relation (BTFR)(Lelli et al. 2017), demonstrates that galactic kinematics are deeply coupled to the distribution of baryonic matter.

1.3 Gas Based Models

Sancisi and others have pointed out the puzzling connection of gas density to the rotation curves. In these gas models, gas is scaled by a factor of about 7(Hoekstra et al. 2001). The Hoekstra method of simply multiplying the gas mass by a number doesn't work in general, and as more and better data came along, such as the SPARC dataset, it was put aside.

1.4 Our Model

Rather than approaching this problem from the bottom up by inventing novel particle physics models or modifying gravitational field equations, this paper approaches the dark mass problem from the opposite direction: top-down phenomenological mapping. Historically, when fundamental microphysics are unknown, progress can be driven by identifying the exact geometric and empirical laws mandated by observation - much as Kepler's laws mapping planetary orbits paved the structural path for Newton's universal gravitation. Thus, we largely avoid investigating the 'why' of our dark matter model here.

We propose a dark matter (called dark mass in this paper) density profile governed by a universal geometric scaling relation: $\rho_{\rm dm} = P/c^3 \cdot n_b^{2/3}$, where n_b is the local volumetric particle density and P is our new universal constant, in Watts, (by dimensional analysis). The $2/3$ exponent is what sets this model apart. With our value for P, a gas with a particle number density of $10^{-5} cm^{-3}$ would have a

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dark matter density of about 150 times that of the gas itself, while at $100cm^{-3}$ the dark mass only adds about 50% to the density of the gas. For everyday solids like water, this formula would add about 0.1g to a cubic metre. We assume that this dark mass has a negligible effect on individual particle dynamics.

2 DATA AND METHODOLOGY

We perform 3 runs of our model, each run is over all 175 galaxies.

- A fit to determine the best power P, assuming that the gas in SPARC dataset is the only gas in the galaxy.
- A fit where a 'core gas' is added to each galaxy, modelling some (likely) missing gas.
  - The same core gas density is added to all galaxies (one more parameter)
  - A best fit core gas density is added to each galaxy (175 more parameters)
- The core gas fits have a circumgalactic medium (CGM) gas added, which models the dark matter halo.

2.1 General outline of the fits

We determine the acceleration due to gravity at each R_{obs} radius of the SPARC dataset from all baryons with the addition of dark mass from our model. The chosen method uses a 3D volumetric approach where each galaxy is modelled in a cylindrical system with a few million points per galaxy. Note that since the velocity contributions from stars are not modified, the 3D volume model is only used to store a particle density in each cell. Each cell then has an X, Y, Z position, a density of gas particles, and a volume. The gravitational attraction for each of the radial points in the SPARC dataset is determined by summing the millions of individual cell masses at their distance from each radial point. This method thus covers induced acceleration from matter at radii beyond the radial measuring point. Each galaxy is modelled as a disk with thickness as calculated using the same formula as the SPARC 2016(Lelli et al. 2016): an exponential vertical distribution with scale half height $z_d = 0.196R_d^{0.633}$. The bulge is ignored for the SPARC gas. Any gas added is done in a spherically symmetric way. The 3D model is large enough (1.1 times R_{max}) so that any gas - wether in or outside of the disk will contribute to acceleration and mass.

Since our model requires a gas density, and the SPARC dataset only has a gas velocity, a small program was written to determine plausible densities, given the V_{gas} data in SPARC. The problem is under defined, so many solutions are possible. We required the gas distribution to be as smooth as possible while still recreating the V_{gas} from SPARC (which itself was created by analysis of gas density measurements). In run 0 we plotted this gas as a velocity for each galaxy to verify that our calculated gas densities were reasonable.

2.2 Error and χ^2 fitting

The SPARC data include errors in the observed rotation velocities, but the errors in the other parameters are unknown and uncertain. For instance errors in inclination directly affect the reported V_{obs}, and gas density errors effect the V_{gas}, etc. In (Li et al. 2018) , the authors fit the 175 SPARC galaxies individually, with up to 4 free parameters per galaxy. Many of the best fit parameters such as inclination are a factor of 2 or more away from accepted values. This is an indication of the uncertainty in the physical layout of our galactic neighbours.

To capture this uncertainty, we assume a 15% error in the calculated V_{tot}. If an error is not assumed fitting algorithims will pour their effort into minimizing the error on a few of the galaxies that have a large number of highly accurate V_{obs}.

In addition to this χ^2 fitting algorithm, other fitting measures were tested, but the overall results only changed best fit outputs by a small amount.

Error results are shown as per point only, as the the measurements are highly correlated. Due to the exploratory nature of this work, a full Bayesian analysis is outside the scope.

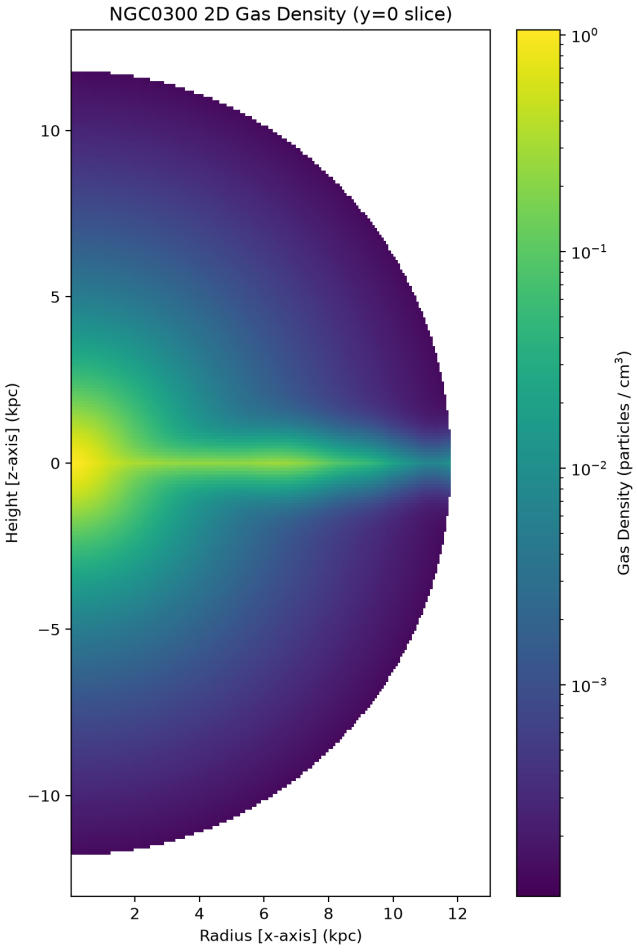


Figure 1. Gas density plot for NGC 300. The galaxy has an additional $0.9cm^{-3}$ gas particles added at the core, plus a CGM gas halo. The CGM halo extends far beyond the diameter shown here.

2.3 Run 0: Zero parameters, Newton

With the SPARC gas recreated as a density, we ran our model over all 175 galaxies. As this run completely lacks dark matter or MOND, it serves as a baseline.

Result:

- χ^2_{Newton} = 578 per point.

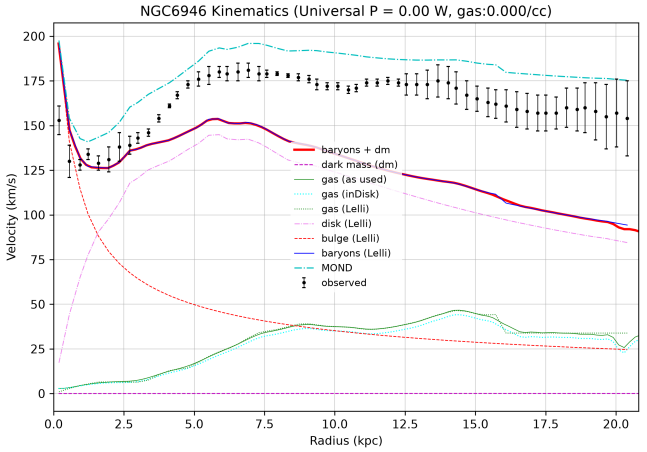


Figure 2. NGC 6946 - Newton. No gas added, our model turned off. This is an example of a check on our gas density recreation. The "baryons + dm" line is directly on top of the blue 'baryons (Lelli)' line which is our test.

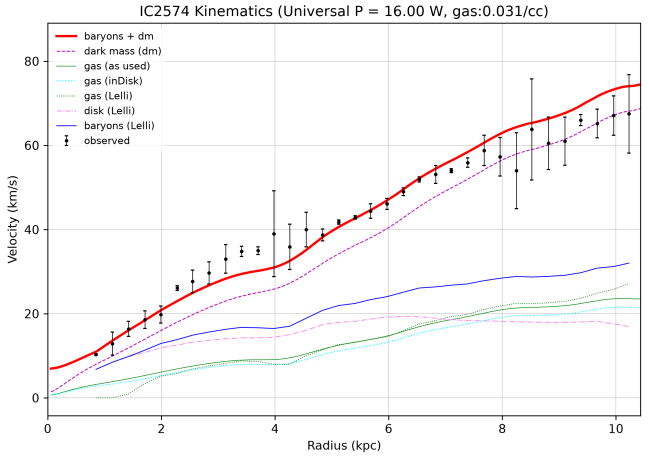


Figure 3. Example individual one parameter per galaxy fit. The only fit parameter 'P' is fixed for all 175 galaxies once it is determined. The purple line shows the total fit by this model. All 175 images are available online. Renzo's rule is honoured at 4 kpc in the IC2574 plot. Note that IC2574 requires almost zero extra core gas for an excellent fit.

2.4 Run 1: 1 global parameter, fit P

We obtainied a dark mass for each cell by using our model on the supplied SPARC gas. The entire 175 galaxy set, with 3319 data points was fitted to obtain a best power.

Result:

- best fit power: 36 Watts,
- χ^2_{Power} = 179 per point.

2.5 Run 2: 2 global parameters, fit P, central density

The SPARC dataset only includes HI gas measurements. There are other sources of gas in a galaxy. In addition, the centres of many galaxies are opaque to some gas measurements(Saintonge & Catinella 2022), so the gas composition and densities at the cores of galaxies are not well known, especially for the purposes of our model, where hidden gas in the galactic core cannot be subsumed into a minor dynamical change in the stellar luminosity parameter, Σ_{bulge} (Lelli et al. 2016).

In order to add a gas core to every galaxy in the dataset, we needed to scale the added gas to the mass and/or size of the galaxy. The additional gas is added via this formula:

$$n_{\rm core}(r) = n_0 \left[1 + \left(\frac{r}{f\, r_s}\right)^2\right]^{-\beta}. \tag{1}$$

Where r is the spherical radius from the galactic centre, n_0 is the central particle density, $\beta = 2.25$, $f = 2.5$, and $M_{\rm gal} = \left(L_{[3.6]}/L_\odot + M_{\rm HI}/M_\odot\right) M_\odot$ and:

$$r_s = 2\,{\rm kpc}\left(\frac{M_{\rm gal}}{10^{11}\,M_\odot}\right)^{1/3}. \tag{2}$$

Note: $L_{[3.6]}$ and $M_{\rm HI}$ are directly from SPARC database columns.

The only free parameter is n_0. In this fit, a single best n_0 was found by minimizing the error while at the same time maximizing the power, P. A larger P is better as it requires les s gas. A power of 16 Watts was chosen.

Result:

- chosen power: 16 Watts,
- chosen core gas beta: 2.25,
- chosen core gas factor f: 2.5,
- best fit n_0: $0.55 cm^{-3}$,
- $\chi^2_{core\ gas}$ = 81 per point.

2.6 The MOND fit

The MOND fit in this work is a very simple implementation: Using the MOND formula(Famaey & Binney 2005) with Lelli gas only:

$$g_N = \frac{V^2_{\rm disk} + V^2_{\rm bul} + V_{\rm gas}|V_{\rm gas}|}{R}, \tag{3}$$

$$g_{\rm MOND} = \frac{g_N + \sqrt{g_N^2 + 4 g_N a_0}}{2}, \tag{4}$$

$$V_{\rm MOND} = \sqrt{g_{\rm MOND}\, R}, \qquad a_0 = 1.2 \times 10^{-10}\ {\rm m\,s^{-2}}. \tag{5}$$

Result:

- χ^2_{MOND} = 50 per point.

2.7 Run 3: a core gas for each.

In fit 3, we ran an optimizer on each galaxy, determining the best fit for n_0 for each of the 175 galaxies individually. The physical argument for adding 175 parameters to the fit, is that the unknown gas in a galaxy will naturally vary between individual galaxies. Of course more parameters could be varied as well. The same gas function was used as in run 2.

- chosen power: 16 Watts,
- chosen core gas beta: 2.25,
- chosen core gas factor f: 2.5,
- best fit n_0: $\in [0, 6] cm^{-3}$,
- $\chi^2_{each\ gas}$ = 14 per point.

We run our python program with SPARC data as given, then adding dark mass as a function of the SPARC gas density.

2.8 The Universal Scaling Model

The local dark matter density ($\rho_{\rm dm}$) is assumed to follow a geometric power-law relation:

$$\rho_{\rm dm} = P/c^3 \cdot n_b^{2/3}, \tag{6}$$

where n_b represents the local volumetric baryon density (calculated here from the raw SPARC neutral hydrogen gas profiles), and P is a universal invariant constant across all systems, determined by a best fit to the SPARC data. Our best fit of the parameter P is:

$$P \approx 59\ Watts. \tag{7}$$

2.9 Statistical Optimization via Least Absolute Deviations (LAD)

Because galactic rotation curve data points exhibit localized spatial dependencies (e.g., radial proximity within individual galaxies sharing systematic streaming motions or beam smearing), traditional ordinary least squares (χ^2) approaches can be disproportionately skewed by local outliers. We instead evaluate the goodness-of-fit using the Least Absolute Deviations (LAD) framework.

The global model parameters are optimized by directly minimizing the total absolute residual sum of the observed ($V_{\rm obs}$) and predicted ($V_{\rm pred}$) rotation velocities across the entire ensemble of $N = 3,391$ data points:

$$\text{Total LAD} = \sum_{i=1}^{N} |V_{{\rm obs},i} - V_{{\rm pred},i}| \tag{8}$$

The resulting model fit is characterized via the Mean Absolute Error (MAE), defined as MAE = Total LAD$/N$, providing a, velocity-space measure of global model precision in units of km/s.

3 PYTHON CODE DESCRIPTION

The raw data is read in from the SPARC website publically availble data, blah blah

4 RESULTS

5 DISCUSSION

5.1 Newtonian Dynamics

5.2 The interpretation of Power

Power is about 10W. If a single baryon were literally emitting or processing 10 Watts of power, it would burn through its own rest mass in a fraction of a nanosecond. Therefore, P cannot be a literal power output of an individual particle. It could be a steady state relationship (as it relies on neighboring particles. Or, Power might represent a coupling constant of baryons to the vacuum or spacetime itself. It defines a linear mass density scale ($P/c^3 \approx 10^{-24}$ kg/m) that dictates how baryonic mass curves space beyond its Newtonian boundary.

5.3 Model Results

The global optimization of our one-parameter model across all SPARC observations, results in a MAE of 26 km/s. The MOND framework as implemented here scores (MAE = 19 km/s). Our model achieves a remarkably similar accuracy profile using only a single universal parameter, like MOND. But unlike MOND, our model does not require the local particles to 'know' the global acceleration parameter, rather, they just have to 'know' the density of the material they are in.

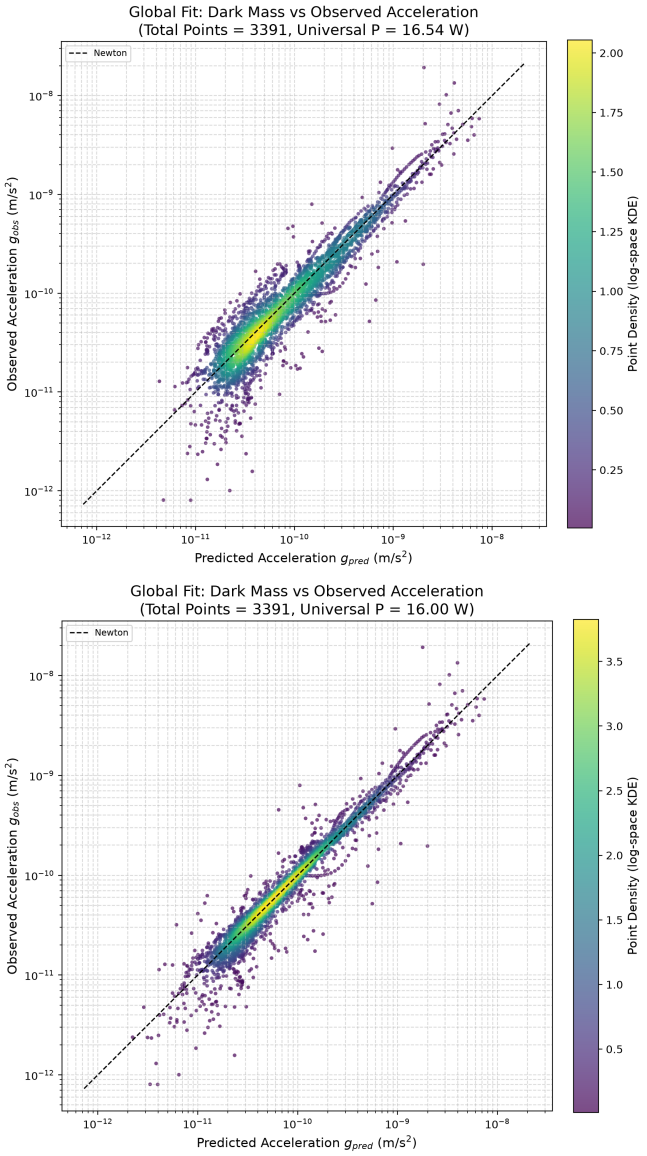


Figure 4. All 3319 SPARC data points are plotted. The acceleration implied by the observed velocity V_{obs} is on the vertical axis, the x axis has the full acceleration from baryons + the dm field introduced in this paper. The red line is a Newtonian expectation, the blue line is a (log) straight line fit to the data. Also: velocity residuals.

5.4 Model Variations

We also ran the model by adding 175 more fitting parameters, allowing each galaxy fit to adjust the entire gas density up or down by up to a factor of 5. This resulted in an (MAE = 17 km/s).

In several SPARC radial bins, central core gas densities are recorded as zero. To test sensitivity to this, we implemented a physically motivated observational detection floor of 0.45 baryons/cm^3

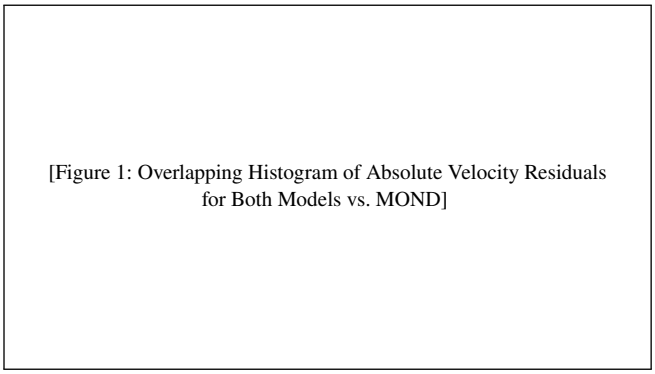


Figure 5. Distribution of absolute velocity residuals $|V_{\rm obs} - V_{\rm pred}|$ evaluated across the 3,391 data points of the SPARC dataset. Vertical dashed lines point out the global MAEs.

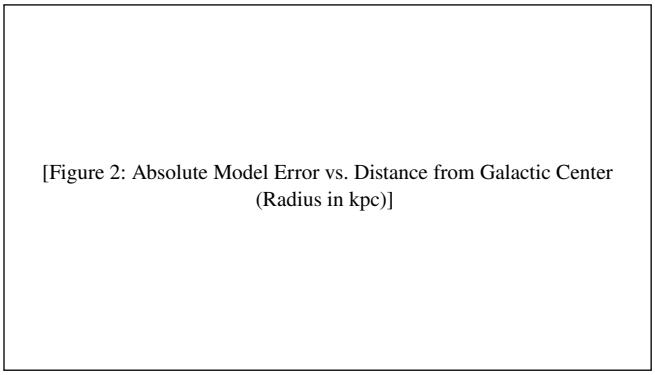


Figure 6. Moving average trend line showing absolute model errors as a function of radial distance from galactic centers, indicating where core fluctuations stabilize.

for unrecorded or low gas densities. Enforcing this floor shifts the global MAE by a negligible fraction, but it does make the global fit line up with Newton extremely well.

These two model variations were not used in any of the other discussions in this paper, full results for them and a few others are available online.

5.5 Core Dynamics and Distribution Tails

An analysis of the absolute residual distributions reveals that MOND's localized advantage is concentrated almost entirely in a tight low-error spike between 0 and 15 km/s, corresponding to high-density inner galactic cores. This divergence is physically expected; our model evaluates dark mass generation based purely on the gas density profiles, which frequently drop or fluctuate in stellar-bulge dominated cores. Crucially, in the intermediate and long-range tails (25 to 100 km/s) - representing the low-acceleration, outer-disk environments - the error distributions of our model and MOND track one another identically. This strongly implies that in the highly rarefied outer halo regime, our geometric density relation functions as a direct mathematical dual to MOND's physical behaviour, without requiring complex acceleration interpolations.

6 CONCLUSION

In this paper, we have evaluated a universal, one-parameter empirical dark matter density profile defined by the geometric relation $\rho_{\rm dm} = P/c^3 \cdot n_b^{2/3}$ against the complete SPARC database. By optimizing the global constant via robust Least Absolute Deviations (LAD) regression across 3,391 data points from 175 morphologically diverse galaxies, our model achieves a global Mean Absolute Error (MAE) of 26 km/s. This performance is comparable to the MOND framework (MAE = 18.7 km/s). Both frameworks use a single parameter to model DM across all galaxies.

The statistical parsimony of this model provides compelling evidence for a universal coupling between baryonic density and dark mass distribution. We demonstrate that the $2/3$ scaling law captures the observations of the dataset.

An analysis of the absolute residual distributions reveals that while MOND maintains an advantage in dense, stellar-dominated galactic cores, the two models track one another identically across the low-acceleration, outer-disk environments. This suggests that in the highly rarefied, widely spaced gas structures of the outer halo, our geometric density relation functions as a direct mathematical dual to MOND's phenomenological behaviour.

By focusing strictly on a data-driven, observational approach, this work side-steps the unconstrained parameter space of particle physics models that have remained undetected across decades of microphysical searches. We do not propose a specific microphysical mechanism for the $2/3$ scaling law, nor do we speculate on the fundamental properties of a constituent particle. Instead, we present this universal relation as an empirical reality mandated by galactic kinematics. We leave it to the particle physics and theoretical communities to engineer a fundamental mechanism—whether rooted in environmental screening, field-baryon interactions, or quantum vacuum geometries—capable of reproducing the strict observational constraints demonstrated here.

DATA AVAILABILITY

The SPARC data underlying this article are publicly available from the SPARC database (Lelli et al. 2016).

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